Python Quiz - Manual Style Answers

BEGINNER

1. What is Python?

Answer: Python is a high-level, interpreted programming language that's easy to read and write. It's often used for web development, automation, data analysis, and more. It supports different programming styles like object-oriented and functional programming.

2. Explain the difference between list and tuple.

Answer: A list is mutable, meaning you can change, add, or remove elements. Tuples are immutable, so once you create them, you can't modify their contents. Lists use square brackets [ ], and tuples use parentheses ( ).

INTERMEDIATE

1. What are decorators in Python?

Answer: Decorators are special functions that modify the behavior of other functions without changing their code. You place a decorator above a function using the @ symbol. They're useful for things like logging, access control, or measuring execution time.

2. Explain list comprehension with examples.

Answer: List comprehension is a concise way to create lists using a single line. For example:

squares = [x**2 for x in range(5)]

This creates a list of squares from 0 to 16.

ADVANCED

1. Explain metaclasses in Python.

Answer: A metaclass in Python is a "class of a class." It controls how classes are created. You can use metaclasses to customize class behavior at the time of creation, like automatically adding methods or checking attributes.

2. How does garbage collection work in Python?

Answer: Python uses automatic garbage collection to clean up unused memory. It keeps track of object references and removes objects that are no longer being used. It also uses something called reference counting and a cyclic garbage collector for more complex cases.